Request for Comments (RFC) on the AI Assessment Sandbox Configurator's API

Towards an open and European "Hugging Face for AI Testing" – March 2026

Introduction

This document is a Request for Comments (RFC) on the concept and open API (Application Programming Interface) of the AI Assessment Sandbox Configurator, an open source tool designed to scale the assessment of AI solutions. This two pager provides a summary overview of its objectives and overview of the documentation to be reviewed preferably before the EUSAIR co-creation workshop on April 16 and 17.

The Challenge: Why do we need a Configurator for AI Assessment Sandboxes?

As AI becomes increasingly powerful and pervasive across Europe's economy and society, the number of AI systems requiring formal assessment is growing rapidly. The EU AI Act establishes clear obligations for AI providers to demonstrate trustworthiness of their AI solutions and designates AI Regulatory Sandboxes as a key mechanism for structured dialogue between innovators and competent authorities (art. 57 and 58).

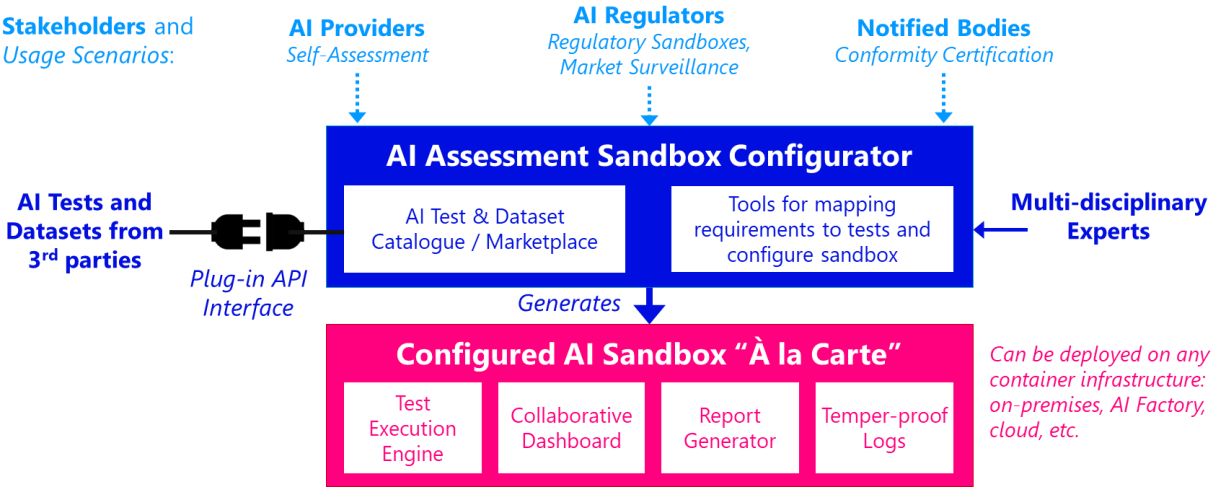
However, a critical bottleneck undermines this ambition: setting up the technical environments required for AI assessment remains highly complex and time-consuming. Converting requirements into metrics, identifying relevant tests from disparate sources, integrating them into a coherent testing environment, collecting and visualising results, generating reports, and maintaining audit trails are largely manual processes — creating a significant barrier to adoption at scale, thus slowing both innovation and regulatory oversight.

The Proposed Solution

To address this bottleneck, the Luxembourg Institute of Science and Technology (LIST) and the University of Luxembourg have developed the **AI Assessment Sandbox Configurator** — an open architecture that streamlines the configuration and deployment of tailored AI assessment environments "à la carte". Incubated within the Luxembourg AI Factory, the Sandbox Configurator supports three stakeholder scenarios:

i. AI Providers conducting technical self-assessment;
ii. AI Competent Authorities operating AI Regulatory Sandboxes and market surveillance; and
iii. Notified Bodies performing conformity certification.

The Sandbox Configurator provides a curated, extensible catalogue of AI tests and datasets integrated through an open plug-in API interface. It guides users through mapping regulatory requirements to assessment metrics, selecting from the catalogue the tests relevant to their specific use case, configuring a collaborative dashboard to execute tests and visualise results, and generating structured assessment reports. Once the AI Assessment Sandbox has been configured, the system generates a bespoke Sandbox that can be deployed on any infrastructure — on-premises, AI Factories, or cloud environments with no vendor lock-in.



Our Vision: An Open Source and European "Hugging Face for AI Testing"

We envision the AI Assessment Sandbox Configurator becoming a European sovereign open-source asset, co-developed by a broad community of stakeholders. As an open architecture, it delivers four strategic benefits:

i. **Scalability**: significantly reduces the time and effort required to configure a tailored AI Assessment Sandbox "à la carte", enabling the extensive trustworthiness testing envisioned under the AI Act across self-assessment, conformity certification, regulatory sandboxes, and market surveillance.

ii. **Curated AI Testing Catalogue:** a structured, extensible catalogue that can become a sort of "Hugging Face for AI Testing" including AI testing tools, controls and datasets integrated through an open plug-in API interface — with the potential to evolve into marketplaces specialized by vertical domain (e.g. healthcare, manufacturing, finance, etc.).

iii. **Common Language**: a shared framework enabling multi-disciplinary collaboration and cross-domain comparability and reproducibility of assessment results.

iv. **Regulatory Learning**: machine-readable AI system cards and assessment exit reports that feed regulatory intelligence at organisational, national, and European levels.

Overview of the Documentation for Your Review

The documentation for your review includes three files on this Luxembourg AI Factory Github Repository:

i) **Concept and Architecture**: this is an overview document describing the context, the definitions, the requirements, the user journey and the open architecture of the Sandbox Configurator. The more technical parts are shaded in grey and can be skipped by non-technical readers.

ii) **Catalogue of AI Tests and Controls**: this document describes the metadata to filter and search AI tests and controls in the Catalogue.

iii) **Plug-in API to integrate third-party tests into the Catalogue**: this is a technical document for developers who wish to integrate existing AI tests into the Catalogue using the plug-in API.

Please send you comments via the following web form: https://tally.so/r/WOPJ0J. For any questions, please contact: alessio.buscemi@list.lu and daniele.pagani@list.lu.

Thank you in advance for your interest and feedback !

--The Luxembourg AI Factory Sandbox Team